

The experience of value and the value of experience: land valuation in the 19th and 20th century.

Introduction: valuation as measurement of value

In 1627, a group of Florentine friends submitted Galileo Galilei an apparently trivial problem that gave rise to a dispute: “A horse is actually worth 100 scudi: by one appraiser the horse is appraised 1000 scudi and by another 10 scudi: it is asked who estimated better and who deviated more from the correct estimate?” Galilei, after some hesitations, answered that the two estimates were equally wrong. Galileo’s opponent in this dispute, Tolomeo Nozzolini, argued instead that the former estimate of 1000 scudi was evidently further from the truth.

Nozzolini took the difference as the right measure of the quality of appraisal. Galilei ridiculed this approach, because it could lead one to consider negative estimates better than positive ones, although appraised value was always a positive quantity. In response to Galilei, Nozzolini claimed that the best judges of similar problems were butchers at the market. They would never agree with the respectable physicist that a loss of 900 scudi was the same as a loss of 90. (Galilei 1896, 576) Galilei riposted that measurement errors were not to be measured in scudi, but as a proportion of the real value:

“A measure used to measure deviations (*stravaganza*) is a deviation itself, and not a scudo, a pound, a rod... one cannot judge the deviation of estimates in any way without relating them to the true value of the thing being estimated.” (Galilei 1896, 588–89)

Despite the acclaim of historians of science who saw in Galilei's answer the sign of a spreading scientific mentality and an innovative stance in the theory of errors, in this case, it is difficult not to recognize the merit of Nozzolini's argument and his appeal to the judgement of market actors. Key to the understanding of both, though, was the idea that the value of the horse was a fixed quantity that had to be measured – an objective theory of value. (A discussion of this episode and its significance is included in Maifreda 2010)

Value theory has been, especially after Ricardo, central to economic theory. Yet, while economists developed various ways of understanding value – with competing objective and subjective theories – the actual measurement of the value of things was left to the ancillary discipline of valuation, which economists rarely mentioned in their own writings, gesturing instead towards the market as the mechanism that generated valuations. Only in special cases, for instance when economists were asked to provide a scientific justification for values in court, did economists actually engage with the practicalities of measuring value. Historians of economics have not been indifferent to these debates and have extensively studied pricing in financial economics (Poitras 2025), the valuation of life (Charpentier and Cherrier 2019), the value of public goods (Maas 2021; Maas and Svorenčík 2017) and of women's work (Chassonnery-Zaïgouche 2023). Nevertheless the history of valuation and pricing has been written mostly by sociologists and historians of accounting (Fontes, Panaretou, and Shakespeare 2024; Chiapello 2015; Racliffe and Munter 1980) and by historians interested in specific mercantile or administrative practices (Bourillon and Vivier 2012; Chauvard, Barbot, and Mocrelli 2010).

In this perspective, agricultural economists of the late nineteenth and early twentieth centuries constituted a very distinctive group. With little institutional link to the rest of economics, and in constant contact with people in the agricultural trade, they did maintain a systematic connection between accounting, valuation, and private and national economics. The study of the valuation of land and farms as it developed in agricultural economics shows therefore the development of a theory of measurement in an intermediate space between theories and practices, between university departments and actual farms.

As Hasok Chang remarked, a (social) constructivist approach to measurement of physical facts presupposes that the historian of science adopts a realist approach to society (Chang 2004). By discussing land valuation, this article tracks the short-circuits that emerge when the scientist and the practitioner are measuring society itself. The literature on social quantification has stressed how measurement creates objects (Desrosières 1993; Porter 1994a) and the performative power of measurement, most notably in the case of the pricing of options (MacKenzie 2006). As Ted Porter summarized the results of this tradition of research, “with numbers one can make new things, or at least transform old ones” (Porter 1994a, 398)

What is notable about land valuation, however, is that people with interests at stake were immediately involved in every individual act of measurement, and could contest the measure taken. In contrast to the statisticians who created distant aggregate measures, appraisers had to take this contestation into account not only in individual practical cases, but also in their own theory of valuation. This article therefore underlines the very real impact of society on measurement and the limitations that society imposed on the power of quantifiers. A judgment about value had to respect the expectations about the real price that

society harbored. For this reason, Nozzolini, in his dispute with Galilei, appealed to the judgment of butchers in the market. They knew better than the physicist how to judge errors in horse prices. I will call this phenomenon, namely the reference to a social world that has its own values, which measures have to respect – *expectationism*.

Expectationism – as a component of measurement - is akin to but not reducible to the appeal to the senses and experience that Chang assumes as the starting point of iterations in his understanding of measure.(Chang 2004, 43) Expectationism implies the reference to highly social expectations and realities, not to the sensorial experience of one individual, but to a life world of discussions, institutions, and negotiations. It is important to notice that valuers may actually challenge, based on their expertise, the drag of expectations, by subverting usual expectations about values. This has been for instance the role played by economists in discussions about the value of women's work (Chassonnery-Zaïgouche 2023) or contingent evaluation (Maas 2021; Maas and Svorenčík 2017). In this case, though, they need to rely even more strongly on the consensus of their professional group.

One of the consequences of *expectationism* in valuation is particularly relevant for economics. In the Hayekian tradition, price acts as the coordinating mechanism *par excellence*, but what is price and how it is formed? A close look at valuation shows that all actors involved in the determination of price have expectations about prices prior to the moment the actual transaction happened and regardless of actual transactions. The theory of valuation, therefore, has to take this into account, on the one hand because the appraised prices have to be realistic (that is, conform to the general expectations in society and to the specific expectations of the other professional valuers in a given context), but also, and

seemingly contradictorily, because actors, including appraisers, considered that market prices could be wrong!

Accountants distinguish various measurement attributes of values. For instance, the framework of the Financial Accounting Standard Boards lists five of them: historical cost, current cost, current market value, net realizable value, and present value of future cash flows.(Fontes, Panaretou, and Shakespeare 2024, 760) In the literature on appraising value appears as an encompassing concept that has to be measured differently in different specific situations. Nevertheless, its key meaning is always the sale price. It is the sale price that appraisers were meant to appraise.

The first two sections describe how the use of valuation as a measure for science - specifically for agricultural economics, emerged out of a much more ancient practice of valuation for trade and practical accounting. For conceptual and institutional reasons – agricultural economists educated professional valuers – the two remained strictly intertwined. Section 1, in particular, also demonstrates the importance of social expectations for the appraising process. Section 3 and 4 explore how the theory of appraisal elaborated competing methods to estimate value and how it conceptualized their anchoring in societal expectations about prices.

Valuation for the economy: the appraisers

It would be impossible to encompass the variety of professional practices linked to valuation in nineteenth and twentieth century Europe: professional norms changed over time and across places, professions, and social groups. The theoretical debates discussed in this article originated mostly from the German-speaking space, characterised by frequent

exchanges and a substantially unity of academic life. Yet, the profession of appraiser across continental Europe was built around broadly similar practices and similar institutions: valuations were required by private individuals for sales and purchases, for credit and insurances (mortgages and foreclosures), or for different legal purposes (including inheritance, trials, etc.), but also by governments for tax purposes, especially in connection with various forms of cadastre derived from Austrian (Milanese) and Napoleonic models (Bourillon and Vivier 2012; Touzery 2007). Indeed, it was in particular the legal and tax requirements that led to a standardization of practices and training and to the emergence of a group of institutionally recognized appraisers, already from the 16th century (Petracco 2016; Petrillat 2007).

In the case of the Sardinian cadaster of the 18th century, for instance, the appraisal of all land parcels throughout Piedmont and Savoy was conducted by government appraisers and surveyors appointed by municipalities according to standardized instructions (at least in principle). While local surveyors were often found lacking either the experience or the ability to carry out the appraisal according to the instructions, it is clear that over time such a massive enterprise shaped professional practices. The cadaster assessed the value of land as the capitalized value of its rent and established therefore a long-standing model for net-income based valuations. Crucial to the valuation was the assessment of the quality of land (its *grade*) based not on the actual land use and profitability but on its potential when run by an ordinary farmer with normal running expenses. It required, therefore, not only information about current product prices, but also, as indicated in the instructions, access to the accounting books of local farmers (those of them who held their accounts!) in order for the appraisers to calculate typical values and expenses. (Savoy 2022)

The professionalization of appraisers and land valuers – including for the standardized needs of government institutions – increased across Europe over the course of the nineteenth century and led to the introduction of appraisal into educational curricula of agricultural and engineering schools at various level, in addition to the publication of handbooks and manuals. Appraisal of farms and land plots was for instance prominent in the teaching of Albrecht Thaer, who was often acknowledged as the founder of agricultural science in nineteenth century Germany. (Uekötter 2010, 43) By the end of the century, in many countries, land valuation was taught together with farm economics and accounting, as showed by the study curricula of Wageningen in the Netherlands (Wageningen), of the Swiss ETH, and of agricultural schools in Italy.(D’Onofrio forthcoming)

The institutional context, together with norms spread by the education system, determined, thus, the general nature of the process of valuation. How where such norms translated into practices though? It is rather uncommon to find self-descriptions of 20th century appraisers, and they have rarely left memoirs concerning their work. The *Erinnerungen* written by Hans Ruedi Hotz after he retired from the Land valuation office of the Swiss Peasant Union in the 1990s are a valuable source.(Hotz 2014) The Swiss Peasant Union ran an accounting office and a land valuation office that worked closely together. The main task of the land valuation office was "the practical implementation of rural inheritance law, particularly in the assessment of productive value," but it also carried out assessments for other purposes, including for perspective buyers. (Agriexpert 2014, 5) I will often use these *Erinnerungen* here to exemplify common characteristics of appraisal and its connection with accounting.

Firstly, there was always a performative element to valuation: the value of a thing, in given contexts, was determined by appraisers. This is easily understood in the context of the land tax, and legal disputes where the report. For instance, Hotz recalls how he was called in to appraise the value of a property in order to determine the compensations for the two heirs. Also market operations, such as mortgages and rents, depended on valuation.(Aereboe 1919, x–xi) All these actions were strictly regulated by laws and the concluding act of the procedure was the appraisal report (*Schaetzungbericht*, in German). In Italy, the continuously reprinted handbook of *Rural, civil, and cadastral appraisal* of the Hoepli publishing house included a model for appraisal reports that ended with the words: “As a result, the definitive value of the farm was calculated in the amount of ... liras” followed by the date and signature of the appraiser.(Ficai 1938, 142) In Switzerland the *Schatzungsbericht* was signed by the appraiser in five copies and delivered to the client, who could always ask for changes and revisions if they were not in agreement with the figures. (Hotz 2014, 12) Appraisal numbers always came with the signatures of the appraiser and the client.

The performative element, in practice, required confidence. Hotz insisted on the physical appearance of the appraiser when he (the gender is intentional) visited farms, probably in the 1950s. For the act of measurement to perform, it had to be carried out appropriately:

“A real appraiser was, of course, male and at least fifty years old. Only an experienced personality was able to make a legitimate and unchallengeable estimate. The appearance of this trustworthy person was of the utmost importance. First of all, it was about the clothing. Rough high shoes, most recently with rubber soles, but once shod with light nails. A pair of sturdy trousers with tight legs that were wrapped with calf bands in winter and wet

weather. A somewhat rough shirt with a tie, a vest and a smock, made of loden fabric in winter and light cotton in summer. The hat was a must. The bag was made of dark leather and was as worn as possible. The age of the bag and the experience of the appraiser had a certain connection.”

Everything in Hotz description is meant to inspire confidence in the appraiser, and specifically in his experience as a practical man, accumulated over a long period of time. We may speculate that experience was not only significant because it translated into a better understanding of the complicated mechanisms of a farm, but also because a long acquaintance with a given place shielded the appraiser from fads and sudden bubbles, that is, it guaranteed that appraised values would be realistic.

Moreover, the appraiser needed tools. There were different kinds of tools. Hotz so described the content of the appraiser’s bag:

“The bag contained the right utensils. It contained the instructions for valuation, the Civil Code and the Code of Obligations (Orell Füssli edition), a schedule of compensations for family labour (Lidlohnbrochure), the Wirz calendar, a folding double metre, a leather pouch with a special measuring tape (*viehketteli*), two blue notebooks (C5 format with squared pages), a slide rule and a wooden box. It contained two pencils. These were sharpened with a multifunction knife. One could be longer, i.e. newer. The other was short and was kept in an aluminium extender so that it could be used up to a length of one centimetre.”

This list includes different kinds of tools: 1) physical measurement tools, indispensable for measuring objects, such as the folding metre or the *viehketteli* to measure the size (and presumptive weight) of cows; 2) technologies for recording and calculating (slide rule,

notebooks, pencils); 3) legal guidelines, such as the civil code and the Lidlohn brochure to calculate the mandated wage of family members; 4) specific instructions. The measurement process consisted in the transformation of physical measures into social values determined by legal and professional conventions. Despite the role of conventions though, the appraisal was assumed to represent an estimate of a “real” value, which was supposed to manifest itself in market transactions – at least, when market conditions were appropriate.

To guarantee the correspondence between the estimated value and the right price, the appraiser relied firstly on his training, on a mix of knowledge of the technical parts of the farm – production processes, plant, soil, animals – and of its economics. A degree in agricultural economics was a necessary but not sufficient condition. According to Hotz recollections, the key site of training for Swiss land valuers was the Renta, the accounting office of the Swiss Peasant Union: it was at Renta that the appraiser gained not only the experience of dealing with farming but also an idea of the actual values of farm production and inventories, before they moved to the valuation office. Renta routinely calculated the return on capital of Swiss farms thanks to the data provided by the valuation office. It also produced the very kind of the data on farms’ net income that were necessary for the calculation of farms’ value according to the Swiss land valuation system (D’Onofrio 2025; Auderset and Moser 2018).

The valuation method prescribed by the instruction and taught in universities was therefore key in the appraisers’ toolbox (or, rather, bag), but the whole point of experience lay in acquiring the skill to estimate realistically, that is correct the results of valuation procedure so that they would come close to the expectations of all parties involved. Hotz for

instance recalls his first appraisal as an apprentice to an experienced employee of the appraisal office. A key step of the procedure was to assign the farm to the appropriate category among the categories established by the accounting office (Bernese clover farm, in this case). This enabled the apprentice to estimate the net income of the farm. Hotz then modified the resulting value to keep track of various peculiarities of the property (the condition of buildings and such things). When Hotz submitted his results to the experienced appraiser, though, the latter corrected them downwards and Hotz noticed: "The man's primary concern was to keep the takeover value within acceptable limits." (Hotz 2014, 11)

Thus, the performative element was kept in check by in check by reference to the common experience of the parties. Experience was therefore not only important as a way of inspiring trust. It held an epistemological value: it guaranteed that the appraiser was actually able to distinguish market noise from real value and keep the appraised value "within acceptable limits". Ultimately this meant adapting the prices that resulted from the dull application of the valuation procedure to reflect the general understanding of the agricultural market. Experience, thus, played a key role in measurements of value in the early twentieth century. It was not only the experience of the appraiser that mattered, but also the experiential world of the other stakeholders and interested parties.

Valuation for science: agricultural economists

If the practices of land valuation had existed for centuries, only over the course of the nineteenth century the theory of appraisal found its place in the emerging constellation of agricultural science, as an ancillary discipline of agricultural economics. According to Friederich Aereboe, one of the most famous agricultural economists in Germany at the

beginning of the twentieth century, “[t]he science of agricultural land valuation and appraisal (Taxations- oder Abschaetzungslehre) is firstly a part of the general science of agriculture and secondly an auxiliary science of the practical science of national economy (Volkslandwirtschaftslehre)” (Aereboe 1919, 1). For him, thus, as for a vast majority of agricultural economists, valuation (of inventories, land, and whole farms) was crucial to farm economics and economic policy because it provided basic measures.

With the so-called Keynesian revolution and the birth of econometric modelling in the 1930s and 1940s, the array of measuring tools available to economists grew dramatically (Morgan 2001; Suzuki 2003b, 2003a), but economists of the 19th and early 20th century, instead, often relied on measures taken *in vivo*, in the living body of the firm. Firm-level data were crucial to their work, as showed by the use of financial statements of joint stock companies to track the business situation, as done by Luigi Einaudi’s group in Turin. (Associazione fra le società italiane per azioni 1912) Agricultural economists, in particular, used accountancy data to understand the connections and relations between the different parts of the farm, the farm and its environment, and the general trend of agriculture. (D’Onofrio 2025) For them, the function of land valuation went beyond the needs of farm management. It was a crucial tool for the determination and assessment of economic policy since it allowed the computation of the return on capital and similar measure of farm profitability. These measures enabled the researcher (or the entrepreneur or the policy maker) to assess the work done, the policies adopted, the management decisions made. (D’Onofrio 2025, 2017; Auderset and Moser 2018)

In order to produce accountancy data, they needed to translate material elements into money measures: valuation operated such translation and involved prices, quantities (such as weights, areas, numbers), and qualities that had to be observed. Valuation thus provided the raw measures of farm value, which were essential for the science of agricultural economics. Valuation became thus itself a key scientific problem for agricultural economists, as witnessed by the massive *bibliography* of land valuation collected by Vittorio Niccoli already in 1889 (Niccoli 1889).

These methodological connections explain why many agricultural economists in Italy and in the German-speaking world wrote four courses: farm economics (Ger. *Betriebswirtschaftslehre*), accounting (*Buchfuehrung/Rechnungswesen*), land valuation (*Taxation*), and agricultural policy (*Agrarpolitik*), as the culmination of the different parts of their science. Agricultural economics was considered a field that included ancillary disciplines (accounting and valuation) and both policy and farm economics. Theodor von der Goltz, who was professor for “Landwirtschaft” in Koenigsberg, Jena, and then Bonn, wrote his handbooks of accounting (1866), valuation (1880), farm economics (1886), general agriculture (1889), and agricultural policy (1899) before turning to agricultural history. Friederich Aereboe, one of Germany’s leading agricultural economists until his retirement in 1931 published a practical introduction to farm accounting (Aereboe 1896), a thick manual of agricultural economics (Aereboe 1917), and a handbook of land valuation (Aereboe 1912), before crowning his career with his handbook of agricultural economic policy (Aereboe 1928). In their teaching, as well as in the curriculum taught at the agricultural university of Wageningen and in Italy, accounting, agricultural economics (both as farm economics and agricultural policy) and land valuation belonged together.

It is unclear how much this “scientific” literature on valuation was based on the formalization of pre-existing practices, but it contributed to homogenize the field. By the late nineteenth century, thus, assessors operating in Europe and North America were educated a relatively standardized sets of methods for the valuation of agricultural businesses and their main asset, namely land. There were essentially two classes of methods of land valuation: the purchase price theory, and the return value method. Agricultural economists and practitioners disagreed whether prices should be based on the market value of land plots and improvements or on the present value of an income flow expected from the land.

These controversies – examined in the following section – demonstrates how, despite a general agreement on the meaning of values, different measurement procedures could arise from different considerations of expediency and concerns for the effect of measurement.

Goltz and income-based methods

Karl Brandt, one of Aereboe’s PhD students in Berlin (and thus a “doctoral grand-child of Goltz), moved to the US in 1933, shortly after the Nazi take-over of Germany. In 1937, he published an article to introduce his American readers to the German debate on land valuation, claiming that “In Germany hardly any other subject within the field of agricultural economics can compete with land valuation as far as the clash between opposing theories and the controversies over the technical approach are concerned”(Brandt 1937). Although farm accounting was equally controversial (with more or less the same opposers), Brandt believed that land valuation was a substantially more important topic, for its crucial practical and scientific implications. Joosep Nou, writing in the 1950s, claimed: “The polemics about methods of valuation (farm appraisal) in the German language sector lasted from the

beginning of the 1910s to the middle of the 1930.”(Nõu 1967, 360–61) The dispute echoed far beyond the German-speaking space, including in the US itself.

This dispute over how best to measure the value of farms and land did not stem from disagreements about the concept of value itself. Agricultural economists, in general, quickly adopted or developed notions of value grounded in scarcity and in laws of supply and demand—well before Alfred Marshall provided a comprehensive reference framework for these ideas.(Sandro 2015, 68) The influence of supply and demand on actual market prices must have been too obvious for them to embrace objective theories of value. In his *Taxationslehre*, Theodor von der Goltz – who perfected the return value method – explained that “The price of a commodity expresses its exchange value in money. It is determined by supply and demand; the less a particular commodity is offered and the more it is in demand, the higher its price, and vice versa.”(Goltz 1892, 25) Aereboe, von der Goltz’ PhD student, but also the most staunch defender of purchase price methods, refined this definition. He distinguished between price and appraised value (*Taxwert*) and stated: “There is always something personal in every value, both with regard to the assessment of the market situation and the assessment of the quality of the goods. An appraised value is nothing more than the expression of an opinion on both” (Aereboe 1919, 95). He claimed that “Value and price match exactly only in the instant the sale is agreed” (Aereboe 1919, 99). Nevertheless, both price and consequently value depended on supply and demand: “it is therefore demand and supply together that condition and determine the prices of goods and the appraisal values to be attributed to them based on these prices”(Aereboe 1919, 98) Goltz and Aereboe’s definitions differ (Aereboe in particular equates value with an opinion about the

market and the goods to be valued), but they stress the close relation between value, price, and supply and demand.

How was then value to be measured concretely? Some functions of the market price of things, that is of their replacement cost or of their sale price, could provide an easy and satisfactory answer for many bits of the farm inventory: tools and machinery, stocks, and buildings. There were complications, though. As Aereboe summarized it, “all valuations of agricultural products will be the more difficult the smaller the relation they hold with the market” (Aereboe 1919, 103). Thus, many issues remained undecided and raised lively debates (how to compute the price of non-marketed products that were reused within the farm, which price was to be imputed to labour paid in kind, etc).

In particular, the appraisal of whole farms or even land plots was not easily amenable to the use of market prices or cost. A first problem was that the land market – and therefore the market for whole farms – was atypical: transactions were few and far between and, as the literature universally remarked, land plots were not easy substitutes for one another. A second problem was more radical: current prices made value contingent on a given place and time and dependent on the particular information available to the parties and their idiosyncratic affections. Agricultural economists often assumed that buyers (at least some buyers) were irrationally attracted to land plots and farms.

Giuseppe Borio, who taught Economics and Land Valuation in Torino since the 1850s (first in the schools of the ministry of finance then in the local university), shared the belief that prices were based on demand and supply, but warned that “The value and therefore the price assigned to a fund, as to any other commodity, by one or a few individuals, either

inexperienced or motivated by love or aversion or any other passion of their own, is naturally variable and uncertain, and therefore unassignable by principles of estimation.” (Borio 1862, 385) Such erratic values should be ignored by the appraiser, who, instead, strived to identify real market value, namely “that which is assigned to the same commodity on the public and free market, where opposing passions are elided, opposing interests are compensated for, and partial judgments are equalized, is, under certain conditions, constant and ascertainable by the estimator. This value, which is said to be current or market value, is that which should be the object of the appraisal.” (385)

Quite quickly, agricultural economists reached the conclusion that such “affection prices” were far from randomly distributed and would not be simply elided by “opposing passions.” As noted for the cases of Italy and Switzerland, they came to believe that small farmers had a tendency to overpay the land and thus distort market values (D’Onofrio 2017). Still in the 1950s, after dramatic change in the political and social structure of the European countryside, the Dutch agricultural economist Geert Minderhoud affirmed, for example: “The land hunger in the Netherlands is so great that countless people are prepared to become farmers, even if they do not earn a reasonable entrepreneurial wage and/or if they do not make a decent interest on their money, while for many money investors agricultural land has a fairly high so-called affection value, which indicates that buyers of land also attach significance to non-business factors.” (Minderhoud 1953, p. 13).

Thus, appraisers ended locked up in a seemingly unresolvable problem: on the one hand, price was the measure of value par excellence and the correspondence between the expected price and the price actually measured in transactions was the proof of the valuation process;

on the other, market prices were recognized by valuers and agricultural economists as unreliable measures of value. Values had to be conceptually insulated from actual market behaviours. For many theoreticians and practitioners, the solution was not to use current prices but, instead, to value farms by a different method.

Historically, indeed, the most widespread methods for the valuation of properties were not based on their market prices. The tradition of cadastral appraising – as we saw – was based instead on the capitalization of the net income of the farm or a similar measure.¹ This practical approach influenced, for instance, by the second half of the eighteenth century the work of the physiocrats.(Alimento, Neri, and Harvoin 2008; Charles and Théré 2012) The fundamental intuition was that the value of the property was the future flow of income it would guarantee to its purchaser.

In Italy and in the Netherlands this kind of methods were called rational or analytical appraisal, since they required a previous detailed analysis of the farm business and knowledge of its balance sheet: the gross product, the costs and operating expenses, the remuneration of capital. As noted by Giancarlo Di Sandro, this approach turned the farm profit and loss account into the foundation of appraisal.(Sandro 1999, 28) In Germany, Albrecht Thaer theorised this kind of methods already in the 1830s.² Von der Goltz, crystallized the net-income method for the German speaking world in his *Taxationslehre* mentioned above (Goltz 1892).

¹ In fact, with time, cadastral appraisal became more sophisticated and the appraiser tried to separate the rent generated by the land itself from that generated by the improvements built upon it and by the entrepreneurial activity of the farmer, but we will ignore this complication here.

² Thaer actually proposed to use cereals, and particularly rye, rather than money as the measuring unit of value, since he believed it to be more stable (Thaer 1833, para. 65 p. 35)

Goltz discussed different cases, but the most relevant for us is the valuation of a whole farm (*Gesamttaxe*, total valuation), consisting of different land plots. The two key steps in the valuation of whole farms were the description and the determination of an “economic plan” of the farm to establish the expected gross revenue and expenses. The description included a crucial piece of information, namely the classification (*Bonitierung*) of the plots to establish their *Ertragsfaehigkeit* or revenue potential. As Goltz explained, the *Bonitierung* could be based on the accounting data of the farm, if those were available. Even when these data existed, though, the appraiser should consider whether the land was currently put to good use. If not, the appraiser should proceed to a new assessment of the revenue potential (*Ertragsfaehigkeit*) of each plot of land.

This was a delicate task, which required experience of local farming habits. Goltz admitted that the laws connecting the natural characteristics of the soil to its gross product (and *a fortiori*, to its net income) were not yet known. Moreover, there was a huge variability of soil composition and of conditions that variously (and almost unpredictably) affected the production of crops and animals. Imperceptible differences caused huge variations in income. Yet, he argued, “Just as the experienced shepherd, from a large flock of sheep—each one looking almost exactly like the other—can nevertheless recognize and distinguish each individual animal by certain characteristics, so too does the skilled appraiser always find differences, even among several plots of land that appear to be entirely the same, which are significant for their yield potential.”

Goltz discussed various ways of classifying the soil based on its gross product per hectare, employed by appraisers in the cadastral administrations of the various German states

(Saxony and Prussia in particular). In general, it seems that Goltz invited appraisers to rely on the different soil quality scales already elaborated by government officials or scientists and adapt to every specific case the corresponding gross product. Goltz claimed, anyways, that it was a good idea to compare the results obtained by the farm's accounts with those estimated by the new *Bonitierung*. If the two methods led to vastly different results, it might be that the land revenue potential was not fully exploited.

The economic plan (or budget), instead, outlined in a simple way the most appropriate use for the fields, buildings and various resources: "The valuation of the whole farm must therefore be based on a uniform economic plan for the asset to be valued, which must show what operating resources are available, how agriculture and livestock breeding as well as any ancillary technical activities are organized, what gross yields can therefore be expected, and what economic costs must be incurred to achieve them." This plan might or might not correspond to the current state of the farm and to the current use of its parts. If the appraiser considered the actual management of the farm to be deficient, he (or less likely, she) could decide to devise a completely new plan, with a new rotation: "A business plan that deviates significantly from the previous one may only be chosen for the purposes of taxation if the previous organization was clearly inappropriate or required an unusual amount of capital and intelligence to implement, and if, at the same time, a better organization can be implemented without significant capital expenditure." In any case, the new plan should be "rational but simple": "Its implementation should only require an average amount of capital and intelligence, i.e., that which is available to men who manage a similarly extensive estate." Thus, according to Goltz' method, the appraiser actually valued not so much a real farm, but a compromise between the actual farm and an imagined average farm. The Italians,

who developed a significantly more sophisticated system of analytical appraisal, spoke of “ordinary farm” (D’Onofrio 2017).

Once the net income was calculated, the value was easily derived by discounting the flow of income by an appropriate rate. Goltz, without any apparent theoretical reflection or justification, suggested local rates of 4 percent for Norther and 3.5 percent for Southern Germany. The difference was meant to accommodate the fact that the high population density in Southern Germany justified high prices – a somewhat circular idea. (Goltz 1892, 464). The result of this – extremely cumbersome – procedure was the appraised value, that is “the actual exchange or sale value of the property at the moment.”

The advantage of appraisal methods based on net income was obvious: since the appropriate market price could not be observed except when the transaction took place, it must be inferred based on available information on the income flow and the discount rate. Theoretically this was easily justified. By 1930, Irving Fisher summarized this point of view by claiming that valuations were determined by expectations about future income: “The value of the good itself is the discounted value of the value (however determined) of its future services.” (Fisher 1930, 16) Yet, there were significant conceptual and practical problems:

1. The calculation of the net income from land was not easy. The farm records did not impute the farmer’s own income and did not divide – how could they? - the income derived from land and from the farmer’s managerial ability or the remuneration of capital (unless capital was borrowed and thus interest was paid). Over the last decades of the nineteenth century, this concept was constantly refined. In the 1910s, the Swiss Ernst Laur, for instance,

proposed to focus instead on a farm's gross income, which was much easier to observe. He assumed, nevertheless, that it was possible to determine statistically a normal ratio between gross and net product of a farm, based on farm types derived from the imposing amount of accountancy data collected by Renta.(Laur 1912b)

2. The introduction of the "ordinary farmer" was not, in itself, a derivation of Marshall's theory of the representative firm. Already in the nineteenth century, appraisers introduced the theoretical construct of the ordinary farmer – which would eventually be used to shift from real, observed farms to "average" farms whose characteristics did not depend on the entrepreneurial qualities of the farmer or the year-to-year vagaries of the weather. But how was an ordinary farm to be identified when accounting records only showed real, existing farms? Again, there was a recourse to generalization/abstraction as the foundation of observation: appraisers did not observe particulars, but organized particulars into generalized observations.

3. Finally, an additional problem with the operationalisation of the net-income method, was the capitalisation rate. The capitalisation rate equalled the ratio of the net-income over the price of the property and thus allowed to calculate the value of the property by projecting into the future the flow of income that the owner expected to receive from it. A low capitalisation rate corresponded to a high price relative to the income. A high capitalisation rate implied a low price relative to income. Even small changes in the capitalization rate produced huge changes in the value. Nevertheless, since the price of a given plot was not observed but when the plot was sold and bought, how could the capitalisation rate be computed? Ideally, agricultural economists recognised that they should

have used the capitalization rate of an equivalent form of investment, for instance government bonds. But in this case, appraisal could exceedingly deviate from market prices. This explains, for instance, why Goltz assumed a lower capitalization rate for Southern Germany: it was not arbitrary, it was a decision based on common knowledge that prices were higher there than in the North for a given level of income. Other agricultural economists and theoreticians of appraisal spent a lot of energy trying to demonstrate why capitalisation rates were so low compared to other forms of investment: the benefits of a healthy life in the countryside, the safety of landed investment, the structure of a market skewed in favour of sellers, were all invoked to explain low capitalisation rates and thus high land prices. (D'Onofrio 2017)

For us, the role of experience is crucial in Goltz's method. The description was entirely based on mature practical experience (*greiste praktische Erfahrung*) and knowledge of local conditions. Ideally, though, not even these two were sufficient. The ideal appraiser or informant was someone who intimately knew the specific farm under appraisal: "The circumstances on which both the gross income and the operating costs of a rural business depend are so varied and often so local in nature that it is impossible for someone who is not thoroughly familiar with the locality to assess them correctly. What net income a soil of a certain quality can yield, how high the wages are, what an average worker can achieve, what kind of cultivation and use of the land is required by the local climate: these and similar things can only be reliably assessed from the outset by someone who has worked in the same area and, if possible, on the same estate for a long time."

The importance of local experience thus explained why local banks, even if their estimation methods were faulty in theory, could still obtain “practically worthy results”: “The reason is also to be found, and not least, in the fact that rural appraisers are always men who are familiar with the local area and well acquainted with local agricultural conditions, who have a sound judgment of the circumstances affecting the yield of the farm or can easily obtain such a judgment.” In practice, theory and method ceded to local knowledge. Goltz’ textbook constantly refers to the experience of the appraiser as a *de facto* foundational criteria.

Thus, income-based valuation methods risked being arbitrary and incurred in circularity: high farm prices though were not inferred from the measurement conducted by the appraisers. They were observed in reality and the rate of capitalization was, instead, adjusted based on “mature practical experience”, that is in order to closely match the locally prevailing prices. Hasok Chang assured that sensory experience is fundamental to the making of thermometers, and we could thus argue that in every district, appraisers had to adjust their thermometer by setting the appropriate capitalization rate based on shared local experience. I underline that, however arbitrary, these methods were not really subjective. Social expectations about the ordinary farmer and about the quality of land existed independently on the appraisers. The appraiser could read such social expectations wrongly and his or her appraisal would be consider invalid.

Aereboe and price-based methods

For this reasons, Friderich Aereboe, proposed to discard income-based methods entirely. His ponderous tome *Die Taxation von Landgütern und Grundstücken*, which came out in 1912 (all citations here are from the second revised edition of 1919), was a frontal attack to income-

based method. Aereboe's critique was directed, at the same time, against the concepts underpinning the theory of net income appraisal, against its economic consequences, and its empirical flaws. As we saw, appraisals were for him just opinions about market prices and values and prices only coincided at the moment the property was sold. In fact, he claimed, properties that generated a similar income flow were often observed to fetch very different prices on the market. Property prices, for him, moved independently from net income (and crop prices), and rose and fall with the general prosperity of a country. As we have seen, net-income based methods partially accounted for these phenomena by adjusting the rate of capitalization, but the result, was for Aereboe that the whole procedure was extremely complicated and, most of all, arbitrary, leaving way too much *Spielraum* to the appraising committee (Aereboe 1919, 316).

He also ascribed pernicious practical effect to the old cadastral assessments, conducted decades earlier, that were still used by banks for calculating the collateral value of farms. By the 1910s they were obsolete and prevented the real appraisal of creditworthiness and risk. The basic classification of soils that underpinned net income methods (the *Bonitierung*) was generally so cumbersome that it could rarely be updated fast enough to keep track of changes in the market price for land and farms. (Aereboe 1919, 317) Thus, Aereboe on the eve of the Great War took issue with net-income methods as proposed by his *doktorvater* von der Goltz or the modified version proposed by his arch-rival Laur, because he claimed that net-income methods were distorting the market for credit. The historian Walther Achilles in the 1990s stressed the political importance of the clash around land valuation in Germany in the wake of the agricultural depression of the 1870s. Lenders insisted so much on capitalisation methods in order to prevent the fall in commodity prices from dragging down land prices

(and thus the creditworthiness of landowners and the banks' balance). (Achilles 1994) In the 1910s, instead, the valuations of the 1880s, based on depressed crop prices, could not keep pace with fast-growing land markets.

Just like Goltz, Aereboe proposed two types of procedures to calculate market prices for farms and land plots. One relied on previous cadastral classifications, for which Aereboe provided updated coefficients that enabled the appraiser to turn net-income classes into prices per hectare. The other instead required the appraiser to fit the appraised property into classes of equivalent price. Both kinds of procedures depended on data about market prices that appraisers had to collect. The latter type also involved a painstaking examination of the quality of the soil and the price of inventories.

In fact, market prices were not so easily observable. Aereboe partly collected price data on his own from sale ads, (Aereboe 1919, 282) but he also relied on a massive data collection of sale prices undertaken by the Prussian government before the First World War. (Aereboe 1919, 280–81) He thus noticed that, in the years around century turn, prices had paradoxically increased more for poorer farms than they had for productive ones. (Aereboe 1919, 282) Moreover, he remarked that land prices did not move in parallel with the prices of crops. (Aereboe 1919, 317) The real difficulty in observing market prices, though, was not simply in collecting raw data. When a farm was sold, it was actually a bundle of things that were sold together. The appraiser should be able to distinguish the price of land from the price paid for building, machines, livestock and the rest of the inventory, otherwise the prices would have been impossible to compare. Aereboe was convinced that it was possible, given the right amount of information, to disentangle the different parts, since the price of

inventories was relatively easy to appraise (at their cost), but these cumbersome calculations inevitably required numerous assumptions (Aereboe 1919, 326) and had then to be averaged over a ten year period in order to insulate real prices from noise.

The data collected and elaborated enabled the creation of an appraisal framework for a given district, that is some basic expectations about usual prices of naked soil and their relation with soil quality, position, and economic factors, in the district. Among the transactions recorded or the sale ads, the appraiser would create a classification starting from the best soils and worst ones and then working out the classes of price in between. (Aereboe 1919, 318–20) The appraiser should then figure out the different factors negatively or positively affecting price, such as distance from markets and cities, the aesthetic quality of the place, the availability of a cheap work force, and others. The framework was thus knowledge “about the location and movement of land properties and land values with which the assessor can and must work (327)” It was knowledge about the price relationship (*Preisverhaeltnisse*) prevailing in the district.

Once the framework completed, the task of the appraiser was thus to decompose the farm under appraisal into its constituting plots and compare each plot, its quality, its characteristics, with the characteristics of plots of land whose prices were known. For instance, they could calculate the average price of land in adjacent areas, modifying it for some special characteristics of the plot to be appraised (superior or inferior quality, connection to roads, presence or absence of water, etc.) in order to estimate the price of the plot.

It is important to underline how the framework was exquisitely local: it established expectations about local prices and about the effect on prices of different characteristics of

the farm (soil quality, natural disposition, and economic characteristics) that were only valid in the specific district of appraisal. Aereboe underlined how the skills of the appraiser were to be acquired through experience, as a progressive acquaintance with local prices and local conditions. This allowed even people with no theoretical knowledge of valuation to be good appraisers: “there are indeed capable agricultural appraisers who achieve good results with relatively little knowledge. This, however, is due to a special skill in translating the extent of existing knowledge into practical application, which in turn is rooted in a highly developed sense of the degree of importance of individual things and moments” (327) To the reader, he made the example of horse merchants – just like Nozzolini had evoked butchers in 1627. Horse merchants may not pass an exam in the theory of valuation, but they know very well what a horse is worth – he observed.

The task of theory then was somewhat to polish and clarify these expectations built on experience: “although it follows from what has been said that land valuation is a skill that grows out of many experiences, it must nevertheless be the task of the theory of valuation to support the acquisition of this experience and skills as much as possible. One cannot learn to ride or swim from books on horsemanship or swimming alone, and yet good literature on these skills is highly beneficial to acquiring them” (Aereboe 1919, 332) In particular, Aereboe expected the academic part of valuation to accumulate data and understand what actually influenced values, but he did not expect the theory to produce a procedure that could be followed mechanically. Theory was “about finding the best way [to appraise value] not the way a blind person can follow in sleep.” (Aereboe 1919, 338)

Ernst Laur reacted vehemently against Aereboe's book, a mere weeks after it came out. In his defense of income-based appraisal methods, Laur ridiculed Aereboe's concerns by claiming he was a georgist, and repeated most of the traditional criticism of price-based valuation methods. To the Swiss, Aereboe was missing the true sense of valuation, focusing instead on floating speculative prices and "Liebhaberwerte". He claimed that price based approaches were thus vulnerable to speculative influences and confused real demand for agricultural land with non-agricultural demands, which Laur viewed as inappropriate for a sound valuation of farms. The result was a disconnect of valuation from actual income-generating capacity and thus from value itself. (Laur 1912a) Laur's own attachment to the (gross-)income capitalisation method revealed concerns with the social consequences of valuation methods, since he wanted to prevent overinvestment in land by small indebted farmers, that would be unable to efficiently run their farms. It was small peasants who drove up land prices at the cost of high indebtedness. The battle between these income-based and price-based appraisal methods was thus a confrontation around diverging political goals and clashing conception of countryside of the future. At the same time, it was a conflict around which expectations had to be taken into account when appraising land, since Aereboe's "market" values also violated some conceptions about real values.

Even Brandt, who was enthusiastic about Aereboe's method, noted, nevertheless its main shortcoming: "It may be seen at a glance that the procedure of appraisal as I have just described it is hardly something that can be practiced by independent individuals. Unless they have at their command most unusual resources, they will not be able to fulfil the preliminary requirements which are necessary to build up a frame for valuation and always to keep it up-to-date. In this respect the capital value appraisal differs fundamentally

from the capitalized revenue method. The latter is essentially individualistic and permits valuation to be stated immediately on any farm. It only requires that the appraiser be able to estimate the possible yields, gross 'returns and costs, which may be an almost unsolvable problem in itself, but does not require any knowledge concerning the level of values or other links of the farm with the environment.”(Brandt 1937, 178) Despite their apparent simplicity, the taxonomy that underpinned price-based classes of equivalence was just as conventional as the imputed expenses in the income-base method. Just like the net-income method, it projected the past into the future to understand the present. What should then an appraiser do, according to Aereboe, in times of fast-changing prices? They should abstain from valuation! In changing conditions, decisions about sale prices had less to do with appraising than with economic forecasting and the individual ability to pay.

Summing it all up

Paradoxically, both classes of methods tried to insulate value from actual market prices. In the case of income-based methods this was done by design. As Aereboe put it, the calculation of net-income value was based on a “schematisierte” gross income, on “schematisierte” costs and a rate of interest that was adjusted “so that the desired interest on the values derived from purchase prices comes out in the end” (316). In the case of market-price methods, though, the isolation was the somewhat contradictory result of various processes, starting with the decomposition of the sale price in the price of land, buildings and the rest of inventory, and ending with the averaging of ten-years prices. The price-based appraisal thus also reflected not momentary fluctuations but past trends, as Laur accusingly

highlighted. As it is often the case in economics, measuring value implied a high degree of abstraction, the construction of a model (Morgan 2001, 237).

On the other hand, what price-based and income-based appraisal methods also shared was a constant reliance on experience. Experience enabled the appraiser to recognize patterns in soil quality, synthesize diverse information, and assess the long-term productivity of plots—much as a shepherd distinguishes their own sheep, as Goltz observed. For Aereboe, experience provided a framework of price relationships. Inevitably, this experience was tied to specific local contexts, given the wide variety of possible relationships between factors and prices. In both cases, the science of appraisal was an effort to organize and discipline an independent body of knowledge about expectations. Appraisers shared this knowledge with a broader localized, community—which included their clients. This imposed clear limits on the freedom of the appraiser and on the performative power of appraisal. Measures had to “blend in” with expectations and fall within a range of acceptable values—this is the “expectationist” character of such measurements.

It may seem that the emphasis on the appraiser’s experience in both classes of methods rendered the judgment of value, like a judgment of taste, entirely subjective. Yet local expectations about prices and values were not arbitrary for the appraiser. On the contrary, the theory of value measurement acknowledged the reality of social price expectations as a problematic starting point. When agricultural economists designed theories of value measurement, or when appraisers had to choose which method to apply, they treated societal structures as constraints they could not alter, not as a space to exert power. Appraisal had to fit into and adjust to expectations about prices and justice that predated it.

Conformity to social expectations was a criterion for choose one method over another, partly because it justified the utility of appraisal as a method for measuring and adjudicating controversies about value—an entity that everyone had some experience of.

The interest of a historical analysis of the theory of appraisal lies in tracing the interlacing of appraisal theory with practices of value measurement. Only by referring to the profession of appraiser and its links with institutions such as the market, the cadastre, inheritance laws, and insurance does the theory of value measurement developed by agricultural economists becomes interpretable. The theory of farm appraisal thus represents not simply an entry point into a crucial element of accounting—valuation in general—but into general problems of quantification and measurement.

Appraisal is not the only discipline in which experience serves as the foundation of judgement. Connoisseurship, for instance, has been proposed as a category to encompass early modern knowledge of art and medicine, where authority rested on experience (Frigo 2012). Yet aesthetic and medical judgements were quintessentially qualitative. Studies of quantification, by contrast, have focused on the creation of classes. Looking at measurements of value, however, shifts attention from the constitution of statistical objects to the activity of assigning individual things to such classes in order to quantify them. The study of the theory of appraisal, thus highlights the complexity of interactions between between the individual experience of the appraiser and the collective experience of the stakeholders involved in the measurement process. In sharp contrast to the conventional history of quantification, which links trust in numbers to the prestige of impersonal objectivity,(Porter 1994b) here we stressed that appraisals came with a signature and depended on local expectations, neither

striving for mechanical objectivity nor properly subjective. If, as Ted Porter claimed, the quantification of social facts appears to create the ontic furniture of the economic world, in this case quantification appears instead as embedded within the social world, which both frames and constrains the act of measurement.

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